

Muhammad Raafey Tariq

Full-Stack AI Software Engineer

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GitHub: github.com/raafey | Portfolio: raafey.github.io | EU Blue Card holder — Eligible to work in Germany

SKILLS

Programming: Python, TypeScript, JavaScript (Node.js), C#, R

Frameworks and Libraries: FastAPI, LangChain, PyTorch, Scikit-learn, Hugging Face, spaCy

AI and LLM: Agentic Architectures, Tool Use, Function Calling, Retrieval-Augmented Generation (RAG), OpenAI API, Anthropic API, Google Gemini

Orchestration: Multi-step Inference Pipelines, Camunda Workflow Engines, RabbitMQ, Knowledge Graphs (Wikidata, SKOS), Vector Search

Observability: Application Insights (Azure), CloudWatch (AWS), Structured Logging, Distributed Tracing, Agent Evaluation Frameworks

Databases: Cosmos DB (vector search), MongoDB, PostgreSQL, MySQL

Cloud and DevOps: Microsoft Azure, Amazon Web Services (AWS), Docker, Terraform (Infrastructure as Code), GCP

CI/CD: Continuous Integration and Continuous Deployment (CI/CD), GitHub Actions, Azure DevOps, Automated Testing, Code Reviews

Frontend: React, Node.js, HTML, CSS

Languages: English (C1), German (A1), Urdu (Native)

PROFESSIONAL EXPERIENCE

Advanced Full-Stack Developer, AI Systems

10.2024 – Present

Vidispine GmbH (Lumine Group), formerly Arvato Systems | Cologne, Germany

- Designed and shipped Vidinet Cognitive Services (VCS), a production AI backend in FastAPI and C# powering MediaPortal for 5 enterprise customers, unifying 7 LLM and vision providers behind one REST contract.
- Built AutoNewsGPT for Al-Jazeera: an agentic GPT pipeline with tool use, a TTS voice analyzer, and the Twelve Labs Pegasus video model, cutting an hour-plus editorial process to ~30 seconds.
- Integrated non-deterministic AI inference into the Vidiflow BPMN engine via AWS Lambda and RabbitMQ, orchestrating 1,000+ LLM and multimodal inference tasks per month without breaking throughput SLAs.
- Led the MediaPortal revamp in TypeScript, React, and Node.js, shipping multilingual metadata management across 24 languages to the European Parliament with a SKOS-based semantic API.
- Implemented observability for AI workflows (structured logging, tracing, latency and cost monitoring via Application Insights and CloudWatch), cutting mean time to debug multi-step LLM failures from hours to minutes.

AI Software Engineer (Working Student)

10.2021 – 10.2024

Arvato Systems, Bertelsmann SE & Co. KGaA | Cologne, Germany

- Built AutoCut GPT (Master's thesis), an agentic FastAPI system where an LLM plans tasks and selects deep-learning models via function calling to generate video highlights from prompts (F1 = 0.96 on task-plan structure).
- Built a production LangChain service in C# using MapReduce and MapRefine for RAG-style processing of documents exceeding 128,000 tokens, unblocking use cases infeasible under standard context limits.
- Replaced a dedicated Azure vector search SKU with a Cosmos DB and Azure Functions service, cutting infrastructure cost from ~\$80/month to ~\$5/month while preserving semantic recall.
- Developed Mediagraph, combining BERT-based NER with Wikidata entity linking to automate metadata enrichment of 1,000+ video assets, removing the bulk of manual tagging.

Data Scientist

02.2021 – 04.2021

Quant Data Solutions | Islamabad, Pakistan

- Built a low-latency order-book imbalance signal engine in C++ with sub-microsecond processing under simulated market loads, plus Python feature engineering to improve regression accuracy on financial time-series.

Associate Software Engineer

06.2020 – 01.2021

AuditXprt Inc | Islamabad, Pakistan

- Owned a document-compliance product for AIA Singapore end-to-end, building OCR-to-classification pipelines via Django REST APIs and adding multi-threaded OCR that cut analysis turnaround from hours to minutes.

EDUCATION

Master of Science in Data Science

04.2021 – 07.2024

Technische Universität Dortmund, Germany | Final grade: 1.9 (German scale)

Bachelor of Science in Computer Science

08.2016 – 07.2020

FAST National University of Computer and Emerging Sciences, Pakistan | CGPA: 3.5 / 4.0